Scrapy Extraction Report

1. Target Website

- **URL:** https://vnexpress.net/the-thao (VNExpress Sports Section)
- **Goal:** Scrape the title, summary (description), and URL for each article listed on the main page and subsequent paginated pages.

2. Method of Extraction

The method used is the **Scrapy framework** in Python, leveraging **CSS Selectors** for data extraction.

Data Field	CSS Selector Used	Explanation
Article Container	article.item-news	This selector targets the main <article> tag, which serves as the block for a single news item on the VNExpress listing page.
Title	h3.title-news a::text	Selects the text content of the <a> tag nested inside the <h3> with class title-news.
URL (Link)	h3.title-news a::attr(href)	Selects the href attribute (the article link) of the same <a> tag.
Description	p.description a::text	Selects the text content of the link inside the <p> tag with class description, which typically holds the article summary.
Pagination	a[data-medium="Next"]::attr(href)	Targets the 'Next' button or link to automatically move to the next page of articles.

3. Sample Extracted Data

When the spider runs, it will produce output similar to the following, ensuring all articles found

on the page are scraped:

```
[
  {
    "title": "Arsenal đứng đầu bảng sau trận thắng sít sao",
    "description": "Thầy trò Mikel Arteta đánh bại Brentford 1-0 nhờ bàn duy nhất của Kai Havertz, qua đó vươn lên chiếm đỉnh bảng Ngoại hạng Anh.",
    "url": "[https://vnexpress.net/arsenal-dung-dau-bang-sau-tran-thang-sit-sao-4700000.html](https://vnexpress.net/arsenal-dung-dau-bang-sau-tran-thang-sit-sao-4700000.html)"
  },
  {
    "title": "Tuyển Việt Nam tập luyện tại Hà Nội",
    "description": "Sau khi trở về từ Iraq, đội tuyển Việt Nam hội quân trở lại Hà Nội để tập luyện, chuẩn bị cho vòng loại thứ ba World Cup 2026.",
    "url": "[https://vnexpress.net/tuyen-viet-nam-tap-luyen-tai-ha-noi-4700001.html](https://vnexpress.net/tuyen-viet-nam-tap-luyen-tai-ha-noi-4700001.html)"
  },
  // ... and so on for all articles ...
]
```

4. How to Run the Spider and Extract to JSON

Prerequisites: Must have Python and Scrapy installed (pip install scrapy).

1. **Create a Scrapy Project:**
   scrapy startproject vnexpress_scraper

2. **Move the Spider File:** Save the vnexpress_spider.py file into the vnexpress_scraper/vnexpress_scraper/spiders/ directory.
3. **Run the Scraper (Extract to JSON):** Navigate to the root of your project (vnexpress_scraper/) and run the following command. The -o articles.json flag tells Scrapy to output the scraped data into a file named articles.json.
   scrapy crawl vnexpress_sports -o articles.json -s FEED_EXPORT_ENCODING='utf-8'

   NOTE: The -s FEED_EXPORT_ENCODING='utf-8' is **CRUCIAL** for correctly exporting the Vietnamese (non-ASCII) characters. Without it, the text will be unreadable or cause encoding errors.